

2017 H2 Q11

Section: Particles and Waves

Topic: Refraction analogy

Question: A car moving from a smooth road to sand is suggested as an analogy for light refraction. Comment.

The analogy has some useful features:

- The car slows down when it enters sand, just as light slows down when entering a denser medium.
- Because one wheel enters the sand first, the car changes direction — similar to a light ray changing direction when crossing into a new medium at an angle.

But there are also limitations:

- The car comes to a halt in deep sand, while light continues travelling through the medium.
- The slowing of the car is due to friction, while light slows because of its interaction with the medium (polarisation/electric field effects).
- Light refraction obeys Snell's Law ($n_1 \sin \theta_1 = n_2 \sin \theta_2$), which the car analogy cannot capture quantitatively.

Revision tips

- Analogies can be useful to introduce a concept, but always note the differences.

- Refraction = change in wave speed → change in direction at boundary, described by Snell's Law.
- Key is that frequency remains constant when light changes medium.